

Swelling

(Edema)

By **Andrea D. Thompson**, MD, PhD, University of Michigan;

Michael J. Shea, MD, Michigan Medicine at the University of Michigan

Reviewed/Revised Aug 2024

Swelling is due to excess fluid in the tissues. The fluid is predominantly water.

Swelling may be widespread or confined to a single limb or part of a limb. Swelling is often in the feet and lower legs. However, people who are required to remain in bed for extended periods (bed rest) sometimes develop swelling in the buttocks, genitals, and back of the thighs. Women who lie on only one side may develop swelling in the breast they lie on. Rarely, a hand or an arm swells.

Leg Swelling (Edema)



This person has swelling in both feet and lower legs.

PETER SKINNER/SCIENCE PHOTO LIBRARY

Sometimes a limb suddenly swells. More often, swelling develops slowly, beginning with weight gain, puffy eyes upon awakening in the morning, and tight shoes at the

end of the day. Swelling may develop so gradually that people do not notice until the swelling is considerable. Sometimes people have a feeling of tightness or fullness. Other symptoms may be present depending on the cause of the swelling and may include shortness of breath or pain in the affected limb.

Causes of Swelling

Swelling that occurs throughout the body has different causes than swelling that is confined to a single limb or part of a limb.

Widespread swelling is most commonly caused by

- [Heart failure](#)
- [Liver failure](#)
- Kidney disorders (especially [nephrotic syndrome](#))

These disorders all cause fluid retention, which is the cause of the swelling.

Another cause of swelling of the lower legs is pooling of blood in the legs. Many middle-aged or older adults, particularly if they have overweight or obesity, normally have a small amount of swelling at the end of the day due to blood pooling. This swelling typically goes away by morning. Blood can also pool in the legs if the valves in the veins are widened or damaged ([chronic venous insufficiency](#)) such as may occur in people who previously had blood clots in the legs. In such people, the swelling usually does not go away overnight.

Many women normally have some swelling during the later stages of pregnancy. However, pregnant women who have a large amount of swelling, particularly if swelling also involves the hands and face and is accompanied by high blood pressure, may have [preeclampsia](#), which can be dangerous.

Swelling that is confined to a single limb or part of a limb and not caused by an injury is most commonly caused by

- A blood clot in a deep-lying vein in a limb ([deep vein thrombosis](#))
- Skin infection ([cellulitis](#))

Many disorders increase risk of blood clots in a vein. Most often, these clots develop in a leg vein but sometimes they occur in an arm vein. Blood clots in a vein can be dangerous if the clot breaks off and travels to the lungs, blocking an artery there (called [pulmonary embolism](#)).

Cellulitis typically causes swelling of the skin over only part of a limb. Much less often, an infection deep under the skin or in the muscles can cause the whole limb to swell.

Blockage of a lymph vessel (such as occurs in [lymphedema](#)) is a less common cause. Lymph vessels, which occur throughout the body, help drain fluid from tissues. If a tumor pushes on the lymph vessels or surgery is done to remove some lymph vessels or nodes (for example, when women with [breast cancer](#) have lymph nodes removed from the armpit), a limb can swell. In many tropical countries, certain parasites can block lymph vessels and cause swelling ([lymphatic filariasis](#)).

Sometimes, an allergic reaction causes swelling around areas such as the mouth ([angioedema](#)angioedema). [Angioedema](#) can also be a hereditary disorder in which swelling comes and goes at irregular intervals.

Evaluation of Swelling

Although swelling may seem like a minor irritation, especially if it does not cause discomfort and goes away while a person is sleeping, it can be a symptom of a serious disorder. The following information can help people decide when a doctor's examination is needed and to know what to expect during the evaluation.

Warning signs

In people with swelling, certain symptoms and characteristics are cause for concern. They include

- Sudden onset
- Swelling of only one leg
- Significant pain or tenderness
- Shortness of breath
- Coughing up blood

When to see a doctor

People who have warning signs should see a doctor right away. People without warning signs who have a history of heart, lung, or kidney disease or who are pregnant should see a doctor right away. Other people without warning signs should schedule a doctor's appointment as soon as possible.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause of the swelling and the tests that may need to be done.

Doctors ask about the location and duration of the swelling and presence and degree of pain or discomfort. Women are usually asked whether they are pregnant or whether swelling seems related to menstrual periods. Doctors also ask whether the person has any disorders (for example, heart, liver, or kidney disorders) or takes any medications (for example, [minoxidil](#), nonsteroidal anti-inflammatory drugs, angiotensin-converting enzyme [ACE] inhibitors, or [amlodipine](#) and other calcium channel blockers) that are known to cause swelling. Doctors also ask about the amount of salt used in cooking and at the table because excess salt can make swelling worse, particularly in people with heart or kidney disorders.

Doctors look for symptoms that may indicate the cause of swelling. For example, people with [heart failure](#) may have shortness of breath during exertion or may wake at night with shortness of breath. People with swelling and easy bruising may have a liver disorder, and people who have recently had surgery or a cast on their leg may have deep vein thrombosis.

During the physical examination, doctors pay particular attention to the area of swelling, but they also carefully examine the person for other signs. Doctors listen to the heart and lungs with the stethoscope because swelling may be a sign of a heart disorder.



Some Causes and Features of Swelling

Cause	Common Features*	Tests†
<u>Angioedema</u> (allergic, idiopathic, or hereditary)	Painless swelling, most often affecting the face, lips, and sometimes tongue Sometimes itching or tight sensation Swelling that does not remain indented after being pressed (nonpitting swelling)	Only a doctor's examination
A blood clot in a deep-lying vein in a leg (typically), an arm, or the pelvis (<u>deep vein thrombosis</u>)	Sudden swelling Usually pain, redness or darkness, warmth, and/or tenderness in the affected area If the clot travels and blocks an artery to the lung (<u>pulmonary embolism</u>), usually shortness of breath and sometimes coughing up blood Sometimes in people who have risk factors for blood clots, such as recent surgery, an injury, bed rest, a cast on a leg, hormone therapy, cancer, smoking, or a period of immobility such as a long airplane flight	Ultrasound study Sometimes a blood test to detect blood clots (D-dimer)

